

ALI AKBAR GONDAL

Computer Science Student — Artificial Intelligence & Machine Learning

Pakistan · aliakbargondal.61@gmail.com · [linkedin.com/in/ali-akbar-gondal-309546320](https://www.linkedin.com/in/ali-akbar-gondal-309546320) · github.com/BugFreeAli

PROFILE

Final-year Computer Science student at the University of Central Punjab, focused on Artificial Intelligence and Machine Learning. Comfortable across the full pipeline, from applied projects in computer vision, NLP, and explainable AI, to academic research, including an authored conference paper on reinforcement learning, curious about extending this work into autonomous systems and robotics down the line.

EDUCATION

BS Computer Science

Oct 2023 – Aug 2027 (Expected)

University of Central Punjab, Pakistan

- CGPA: 3.39 / 4.00
- Relevant Coursework: Artificial Intelligence, Design and Analysis of Algorithms, Data Structures and Algorithms, Linear Algebra, Probability and Statistics, Differential Equations

FSc. Pre-Engineering (Intermediate)

2022

Punjab Group of Colleges, Pakistan

- Distinction in Mathematics and Physics

RESEARCH

Research Interests

Reinforcement Learning and Reward Shaping · Advance Controlled Systems · Autonomous Systems & Robotics

Publications

Overcoming Temporal Constraints in Directed-Energy Weapons: A PPO-Based Agentic Approach to Heterogeneous UAV Swarm Defense

23rd International Bhurban Conference on Applied Sciences & Technology (IBCAST) · 2026

- Proposed a PPO-based framework for assigning directed-energy weapon resources across heterogeneous UAV swarms — factoring in threat level, laser dwell time, and turret slew time — which outperformed standard fixed-priority scheduling in simulation.

Learning When to Let Go: Reinforcement Learning for Uncertainty-Aware Payload Separation Timing

Frontiers of Information Technology (FIT) · 2026 — Under Review

- Examines when a reinforcement learning agent should trigger payload separation under sensor and environmental uncertainty.

Ongoing Research

- Investigating fuzzy-logic-driven reward shaping for sparse-reward reinforcement learning
- Studying cooperative multi-robot reinforcement learning under limited inter-agent communication

PROFESSIONAL EXPERIENCE

Data Analyst Intern

Summer 2025

DFT, Gujranwala Branch

- Cleaned and analyzed operational business data to build reporting dashboards for the team.
- Surfaced trends and performance gaps that sped up data-driven decision-making.

SELECTED PROJECTS

Crisis OS — Autonomous Urban Crisis Intelligence Platform

PyTorch · Python · CNN

- Coordinated nine specialized AI agents into one real-time system that detects, verifies, and predicts urban crisis events from live data feeds.

Resume Screening AI

NLP · Transformers · Flask

- Applied NLP to automatically score and rank resumes against job descriptions, cutting down manual screening time.

RetinaLens AI — Clinical Decision Support for Diabetic Retinopathy

Computer Vision · EfficientNetV2 · React

- Built and deployed a computer-vision model that flags diabetic retinopathy from retinal photographs through a fast, secure web interface.

ImageSense — Synthetic Media & Deepfake Detection

tensorflow · XAI · React

- Paired vision-based anomaly detection with explainable AI to flag manipulated media and generate a readable forensic report for each result.

Causal Impact Analyzer

Python · Streamlit · BSTS · Plotly

- Built a Bayesian Structural Time Series tool for estimating the causal effect of an intervention — a launch, a policy change — on a time-series metric.

TECHNICAL SKILLS

- Machine Learning / Deep Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face
- Languages: Python, C++, JavaScript, SQL
- Tools & Platforms: Git, Docker, AWS, Linux

CERTIFICATIONS

Machine Learning Specialization2025

Stanford Online & DeepLearning.AI — Coursera

Data Science Foundations Specialization2025

IBM & University of London — Coursera

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

AI Seekho 2026 — Google Antigravity National Hackathon(Pakistan)

- Built a multi-agent AI prototype for urban crisis response as part of a national hackathon.

Organizing Assistant, Digital Logic Design Exhibition

University of Central Punjab

- Helped coordinate a university-wide exhibition featuring 40+ student projects.

Survey Team Lead, Psychology Department Research Study

University of Central Punjab

- Led a 5-person team through survey design and data collection for a faculty-supervised study, reaching 25+ participants.